

Ludonarrative Hermeneutics: *A Way Out* and the Narrative Paradox

Christian Roth¹, Tom van Nuenen², Hartmut Koenitz¹

¹HKU University of the Arts Utrecht, Nieuwekade 1, 3511 RV Utrecht, The Netherlands

`christian.roth@hku.nl`; `hartmut.koenitz@hku.nl`

² King's College London, WC2R 2LS London, United Kingdom

`tomvannuenen@gmail.com`

Abstract. The practice of designing Interactive Digital Narratives [IDN] is often described as a challenge facing issues such as the “narrative paradox” and avoiding the unintentional creation of “ludonarrative dissonance”. These terms are expressions of a perspective that takes narrative and interactivity as dichotomic ends of a design trajectory, mirroring an enduring discussion in game studies between positions often cast as ludologists and narratologists. The dichotomy of ludo versus narrative is, in itself, problematic and is often the source of the very conflict it describes. In this paper, we investigate this issue through the example of the cooperative game *A Way Out*, in which two players team up to break out of prison. The game is designed with a narrative twist, involving the escalation and final resolution of the game’s competitive motif in the final scene. To understand the user experiences of this reveal, and the concomitant consequences, we engage in a discursive analysis of “Let’s Play” videos as a largely untapped resource for research. By analyzing the interactions and performances in these videos, we can more clearly understand player responses to unsatisfying IDN design. As a result we introduce the notion of a ‘hermeneutic strip’, extending Koenitz’ SPP model to locate and describe the involved processes of narrative cognition in IDN work.

Keywords: Interactive Narrative Design, ludonarrative dissonance, narrative paradox, hermeneutics, hermeneutic strip, SPP Model, role distancing.

1 Introduction

The practice of designing Interactive Digital Narratives [IDN] is often described as a challenge that requires moderation between player freedom and the structured experience that interactive forms like video games internalize. Here, terms such as “ludonarrative dissonance” and “narrative paradox” are often used expressions of a perspective that takes narrative and interactivity as dichotomic ends of a design trajectory, reflecting the enduring trope of the narratology vs. ludology debate. However, as we will point out, the dichotomy of “ludo” versus “narrative” is in itself problematic, and is often the source of the very conflict it describes as a design challenge.

In practice, the terms “ludonarrative dissonance” [1] and “narrative paradox” [2] are used in relation to unsatisfactory user experiences in terms of agency and immersion [3]. Little is known, however, about the exact nature of the underlying issues, of the supposed tension between narrative and interaction. We investigate this issue through the example of Hazelight Studios’ narrative action-adventure game *A Way Out* [4], in which two players team up to break out of prison and escape the authorities. We focus especially on player reactions to the showdown in the final scenes of the game where agency is suddenly revoked and the players are forced into a deadly confrontation from which there is no escape. In order to understand how this reveal and change of gameplay is experienced, we turn to the material of “Let’s Play” videos on YouTube, which we consider a largely untapped resource for user-experience research. By analyzing the interactions and performances in these videos, we can more clearly understand player responses to IDN design.

This paper argues that the concept of ludonarrative dissonance is insufficient in describing situations where games let their players down due to tensions between gameplay and narrative constituents. We go on to suggest that the concept should, at least in this specific case, be extended and better defined by using the construct of *interpretation* of the ludonarrative to better accommodate the hybrid nature of the ludonarrative construct taking both player and game designer into account than the entrenched dichotomic ludology/narratology perspective.

2 Summarizing *A Way Out*

A Way Out is a game about the prison breakout of Leo and Vincent, two inmates who meet in jail and become partners in planning a daring escape. Convicted of robbery, assault and grand theft, Leo is introduced as a confident, pragmatic and headstrong character with a tendency towards violence. Vincent, on the other hand, has been sentenced for fraud, embezzlement and murder; he is portrayed as a smart, rational and reserved character. An unlikely pair, the two are united by their wish to take revenge on a common enemy – Leo’s former boss, Harvey, who has framed Leo and killed Vincent’s brother. Determined to make their escape, the two have to work together. Cooperation is required for most tasks, be it opening doors, subduing guards, hoisting each other up, or creating distractions so the other character can accomplish a task.

Teamwork is at the core of *A Way Out*’s game design, realized as cooperative activity for two players presented in a split screen. This configuration supports the narrative of the two characters being dependent on each other, while placing the respective players in the position of their avatars [5]. As Nitsche [6] argues, players understand the video game space and their movement therein by ways of narrative comprehension: a form of understanding of the events they trigger and encounter. To bolster the game’s synergistic disposition, both characters’ story unfolds simultaneously on the screen, allowing each player to see what the other one is experiencing. In certain situations, the screens merge, marking crucial points of interaction and shared experience. Once the two characters have escaped, action scenes are intercut with slower narrative segments, revealing the gradual development of friendship. Throughout the game players encounter

dichotomic choices, usually juxtaposing Vincent’s careful approach with Leo’s brutish one. The narrative comes to a climax as both players finally confront and kill Harvey at his hideout in Mexico. Having fulfilled their mission, they return to the US, but in typical Hollywood fashion, the game has a twist up its sleeve. Vincent turns out to be an undercover cop whose sole reason for working with Leo is to get to Harvey. Feeling betrayed, Leo takes Vincent hostage, leading to a series of actions that culminate in a final showdown. Here, the game’s dominant mode changes from one of cooperation (where players must help each other in order to proceed) to one of confrontation (where players can and, in the end, *must* harm each other in order to propel the narrative forward and achieve the game’s objectives), mirroring the narrative of the betrayal and the resulting broken bond.

That is not to say that confrontation is absent beforehand: in fact, the game constantly plays off of the dialectic between competition and collaboration to increase the bond between players. From the start of the game, small, individual mini-games can be undertaken, such as doing pull-ups in prison, the results of which can be compared to the other player’s tally. However, in the game’s closing moments, narrative and gameplay shift dramatically from a mode that is mainly co-operative to one that is solely competitive. In the last interactive moment, players fight to reach a single weapon. Similar to earlier friendly competitions, the player who presses X faster gets to the weapon first. Once one of the characters reaches the gun, the game cuts to the perspective of that character within a slow-motion scene. The scene produces a radical shift in what, following Genette [7], we might call ludic mood: the way the designer allows events to unfold. We might say this is where the second betrayal occurs – this time, of the contract between the game and its players [8]. The narrative only advances when the trigger is pressed long enough so that the character in control raises his arm, aims for the other character’s chest and then pulls the trigger. This is the only way to advance the narrative of this scene; any choice of non-participation just causes the game to pause.

3 Theoretical Background

3.1 Harmony, Dissonance and Narrative Paradox

< don’t make me shoot him > ↑ I DON’T WANNA SHOOT HIM ↑

The above line is a Conversation Analysis excerpt from a Let’s Play video of *A Way Out* [9]. The player, shouting at the game as he struggles to avoid killing his accomplice and co-player, is typical for the response of players in the final section of the game, having cooperated with the other player until this point. How can we understand these strong, disapproving responses? We might be tempted to call this a matter of ‘ludonarrative dissonance’, which is typically understood as the result of a disconnect between game mechanics and narrative. The concept was first offered in a rather terse blog post by Clint Hocking [1], who defines the concept loosely through the example of *Bioshock*. In the game, the “ludic contract” between player and game, – an objectivist morality that regards everything around them as a means to their own end – is in conflict

with its “narrative contract”, in which the player only progresses when they help an anonymous interlocutor over the radio. *Bioshock* thus contains a “dissonance between what it is about as a game, and what it is about as a story” [1]. The concept has proven fruitful outside academic circles and is frequently applied in games journalism and public discourse.

In the case of *A Way Out*, if we take the game’s dominant ludic mode to be that of collaboration, then, the narrative component seems incongruent, and – in the final scene – violates the “narrative contract” through the removal of agency, eliciting strong negative reactions.

We argue, however, that “ludonarrative dissonance” does not adequately describe this phenomenon, in part due to the unstable ontological premises of this term. While the term ludonarrative dissonance seems to evoke diachronicity – a tension or clash resulting from the combination of two disharmonious or unsuitable elements during a musical piece – the term very often describes its problem in holistic terms. A game either is ludonarratively dissonant, or it is not. The qualitative statement Hocking makes about his case study (“*Bioshock* seems to suffer from a powerful dissonance between what it is about as a game, and what it is about as a story”) demonstrates this *pars-pro-toto* well. Of course, Hocking’s example is not *entirely* ludonarratively dissonant: he notes that during the initial hours of his playthrough, what he experienced was more like what Pynenburg [10] calls “ludonarrative harmony”, as he writes: “the game literally made me feel a cold detachment from the fate of the Little Sisters, who I assumed could not be saved.” Harmony and dissonance seem, much like in music theory, co-dependent terms. Second, when understood in a particular way, ludonarrative dissonance may constitute what Gilbert Ryle [11] designated as a category-mistake, in which things belonging to a particular category are presented as belonging to a different one. We argue that the “ludic contract” Hocking describes is, upon closer inspection, another narrative contract: the world-making that players engage in through the game’s mechanics is a cognitive achievement, and one in which their identification with “Randian objectivism” is a wholly narrative model, based on connected events. This is no attempt to drag up the problematic dichotomy of narratology versus ludology; instead, we mean this insofar as Jerome Bruner noted that “[w]e seem to have no other way of describing ‘lived time’ save in the form of a narrative.” [12] of course, the player *does* engage with mechanics that are not (overtly) narrative in terms of their practical involvement. A reflection on what those mechanics mean, however, undoubtedly needs to consider the narrative aspect. This means we need to look more closely into the tension, between narrative aspects in a game, instead of stopping short at the definition of “mechanics”. We will do so in the analytical section of this paper.

As mentioned, ludonarrative dissonance also implies ludonarrative harmony. Ludonarrative harmony refers to the successful syncing of both ludic and narrative aspects to build a consistent, immersive experience. When achieved, this harmony results in an internally consistent world that “feels right” [10]. As such, ludonarrative harmony seems to describe the status of the *well-functioning videogame*, and its particularities are interesting to analyze. Interestingly, the lead designer of *A Way Out*, Josef Fares, has implemented cooperative gameplay before in *Brothers: A Tale of Two Sons* [13]. In *Brothers*, the game mechanics require players to use one gamepad stick for each of

the two brothers they are controlling, reflecting the cognitively challenging task of co-operation. The older brother helps the younger one with different tasks – notably, carrying him on his back while swimming. Later in the game the older brother dies, and the younger brother has to finish the journey alone. Players are left with one of the gamepad sticks to interact. Yet at the game’s end, players are confronted with another stream to cross, and the younger brother refuses to go beyond the shore. The problem is overcome when players realize that the older brother is still there ‘in spirit’: both gamepad sticks can still be used. This narrative twist, told through the gameplay gambit of first taking away agency and then giving it back, has brought much praise to Fares and his design team. Similarly, in *A Way Out*, Fares’ design fosters the bond between the characters (and the players) through cooperative gameplay right up to the climax of the game, where agency is taken away. In this case, however, it is never given back. Instead, the narrative design imposes either of two versions of an emotional ending, with one character dead. On an analytical level, the question is whether the observed problem is an instance of ludonarrative dissonance: the incongruence between what players assume based on what they have thus far experienced, and what the game designer has planned as the conclusion for the story.

Cliff Makedonski draws several conclusions from the notion of ludonarrative dissonance, considering designers responsible solely for the disruption. Overcoming the problem would either mean “focusing its efforts in a mostly linear direction”, so that the player cannot make decisions that would put consistency at risk, or creating a proverbial map that spans the entire territory, and “molding a world where anything could happen” [3]. Of course, creating an IDN system that allows for such high levels of agency shifts the conversation to whether the resulting experience would still be meaningful for both player and the designer.

This brings us to the concept of narrative paradox [2, 14] which describes the inherent tension between authorship and participation, in which the player asserts agency, the freedom to take actions, while the game designer refuses to relinquish control of the narrative for the purpose of ensuring what they believe is a satisfying structure [15]. In IDN, interactors are encouraged to actively create belief by performing or, in effect, inhabiting a role [16]. This perspective assumes a fundamental distinction between agency and dramatic structure. A narrative is taken as a carefully woven product the storyteller crafts in order to create maximum impact. Herein lies the problem. Costikyan [17] writes: “To the degree that you make a game more like a story – a controlled, pre-determined experience, with events occurring as the author wishes – you make it a less effective game. To the degree that you make a story more like a game – with alternative paths and outcomes – you make it a less effective story.” However, this view has been challenged by a range of authors, including Jennings [18], Murray [16], Mateas [19] and Koenitz [20]. Thus, in order to be able to identify more precisely the tension caused by the ending of *A Way Out*, we need a better view of the ludonarrative relationship between players and designers. To do so, we need to return to the problem of what ludonarrative refers to in the first place.

3.2 Ludonarrative hermeneutics

Adding to the confusion about ludonarrative dissonance is the fact that “ludonarrative” itself “is variously understood as a structural quality of the video game artifact, an experiential quality during the experience of a video game, or a high-level framework to understand video games.” [21] In either case, as we mentioned in the last section, a number of scholars emphasize the difference to linear narrative manifestations and therefore work towards specific theories of video game narrative, seen as a variant of interactive digital narrative (IDN).

As Montfort [22] and Koenitz [23] clarify, an IDN artefact is not itself a narrative, it is an interactive computer program with the potential of instantiating narratives through user interaction. Koenitz reflects this aspect in his SPP model [20, 23], by understanding IDN as comprised of System, Process, and Product. Through the *process* of the player’s engagement with the interactive narrative *system* by choices and other behavior - her performance (see also Knoller’s perspective on “userly performance” [24]) - a concrete and personal narrative *product* is instantiated. Koenitz understands the IDN system as a prototype for all potentially instantiated narratives and thus as a *protostory*.

At any moment of reflecting on this narrative, it forms a story in the mind of the player. This is the ludonarrative expression of the hermeneutic circle, in which a text as a whole is established by reference to the individual parts and one’s understanding of each individual part by reference to the whole. Yet, in the case of interactive digital narrative, the player also makes plans by evaluating the results of the interaction strategy so far, by speculating about paths not taken and by considering the current potential for interaction.

To illustrate how this system of ludonarrative hermeneutics works, we propose to extend Koenitz’ SPP model [23] by adding a “hermeneutic strip”, incorporating two interconnected loops of narrative interpretation processes (see Figure 1).



Figure 1: Extension of Koenitz’ SPP model [23] through implementation of a double-hermeneutic circle (‘hermeneutic strip’)

The hermeneutic strip illustrates the player’s narrative meaning-making process that happens a) when players interact with the system and b) when they interpret the instantiated narrative at any point of the experience. The latter can be understood as the classic hermeneutic circle, and together the two loops result in an interdependent double-hermeneutic circle of the IDN experience (cf. [25]).

In the ‘upper hermeneutic circle’ players are reflecting what the system allows them to do and what narrative they could extract, also in the sense of narrative game mechanics [26]. Within the ‘bottom hermeneutic circle’, then, players interpret the currently instantiated narrative. While players interact with an IDN system, they are continuously extracting information to understand past and present events and to plan their actions. As we mentioned above, this meaning-making process can be understood from a cognitive perspective on narrative. Following cognitive narratologists Herman [27] and Bordwell [28], Marie-Laure Ryan [29] narrative meaning is a cognitive construct, or mental image, built by the interpreter in response to the narrative construct (text in narratological terms). Intuitively, we might understand narrative as located with a narrative product like a printed book or a movie, however, the cognitive perspective stresses the point that narrative resides within the human mind as a mental construct. “... it does not take a representation proposed as narrative to trigger the cognitive construct that constitutes narrativity” [29]. Through interaction with a game such as *A Way Out*, players create mental narratives in an effort to make sense of the simulated environment and constantly check if their assumptions are consistent. New information leads to an updated projection, and conflicting information can lead to perceived dissonance. Thus, we understand IDN to have different narrative agents: on the one hand what is conveyed, implied, intended by the designer on a system level, and on the other hand what is performed, interpreted, assumed, expected, speculated and planned by the player. For the present discussion, we focus on player reactions to an IDN system. Through an analysis of these responses, we will now attempt to shed light on the potential dissonance players experience with the ending scenes of *A Way Out*.

4 Method: Analysis of Let’s Play Videos

For our analysis of player responses to *A Way Out*’s narrative twist, and the way in which ludonarrative hermeneutics can be demonstrated, we turn to Let’s Play (LPs) videos on the popular video platform Youtube. Let’s Play is a style of videos documenting the playthrough of a video game, usually including commentary by the gamer. Glas et al. [30] identified LPs’ potential for game archiving and exhibition purposes within cultural heritage institutions. Extending this perspective, we use LP as samples for the evaluation of user experiences. The first selective step consisted of watching a number of popular Let’s Play videos (N = 40), showing up under the search query “A Way Out Let’s Play”. We narrowed the selection to 20 videos for the conversational analysis when the responses to the telling moment seemed not to yield any new insights.

For the purpose of this analysis we focus on two ‘telling moments’ at the end of the game, when the narrative twist takes place. In the first scene Vincent reveals that he is an undercover cop and Leo becomes aware of his betrayal. The second scene is the

showdown in which Leo and Vincent fight each other, leaving one of them dead. We decided to exclude Let's Play heavily edited videos as we were interested in a recording situation that resembles an unedited think-aloud protocol. A smaller number of conversations (6) were then transcribed using the *Jefferson system*, one of the founders of conversation analysis (CA). [31, 32]. This method is interesting for several reasons. Reeves et al. [33] note that investigations into video game play developed from ethnomethodological and conversation analytic (EMCA) perspectives remain scarce. EMCA approaches the need to engage both with play as social action and the ways in which it is practically accomplished by players, "between players themselves, and between players and "the game" – as a moment-by-moment, sequentially organized activity" (ibid.) They do, however, offer the opportunity to focus in detail on the embodied act of play itself. Yet, EMCA typically attempts to work with what we could refer to as "naturalistic data": audio and video recording of activities that are, as much as possible, occurring in their original settings.

While the kinds of performative behavior that Let's Players demonstrate are likely not the same kinds of behavior found in everyday – and unrecorded – home scenarios, we contend that the performativity here allows us *more* insight, not less, into the ways in which players respond to the tensions arising out of the particular combination of gameplay and narrative. Youtubers "act out" their responses, often using what Ytreberg [34], following Goffman [35], refers to as a performative aspect of mass media, namely hyper-ritualizations and dramatic scriptings. The former involves a concentration and intensification of certain traits of unmediated behavior and interaction. The latter, Goffman notes, refer to "all strips of depicted personal experience made available for vicarious participation to an audience or readership" [35]. It is this dual nature of the Youtube performance that makes it indicative of the way players "ought" to respond to, in this case, a game. Goffman writes that "[the] deepest significance [of dramatic scriptings] is that they provide a mock-up of everyday life, a put-together script of unscripted social doings, and thus are a source of broad hints concerning the structure of this domain." We view the videos we analyze, then, as "over-performed" interactive narrative user experiences, telling of the ways in which games produce affective responses in their players.

To transcribe the findings of the LP sessions we turn to Conversation Analysis (CA): cf. [31, 32], which focuses on "the shape and form of the ways in which contributions to interaction form a connected series of actions", while attempting to "discover and describe the architecture of this structure: the properties of the ways in which interaction proceeds through activities produced through successive turns" [31]. In the process of annotation, we made use of the "Jefferson system" after its developer, the late Gail Jefferson. One of the additions we decided to make to this system is the inclusion of physical behavior, which due to the video format includes important interactional cues. We further made a distinction between physical player behavior (throwing the controller away) and in-game character behavior (pointing the gun towards something), as the latter is obviously controlled by the player but adds further contextual information.

5 Analysis

As the final confrontation between the characters commences, all of the players remained highly motivated to continue, as this event at first seems in line with the gradual escalation of the competitive elements that it has included from the beginning. Then, after a long fight scene between the characters Leo and Vincent (both badly wounded) stumble to a gun at the edge of the building. Still enthralled by the game's ludonarrative harmony, players shout and curse at each other as their movements bring them closer to the gun. Then, as one player reaches the gun, they are confronted with the results: the camera pans behind the winning character, and all they can do is to press the trigger.

In all of the analyzed LPs in which Vincent was the victor, players did not want to shoot the other, trying out a myriad of ways to avoid the outcome. Some temporarily put down the controller in refusal [9], others started playing meta-games with the mechanism of pointing the gun at the player [36]; yet others simply waited for a long time [37]. A closer look at a particular response of the players in The Let's Play channel *T&J Nexus* [38] demonstrates the disbelief of players as the game forces their hand and negates the ludonarrative harmony of comradery that has characterized the majority of the playthrough.

- P Vincent: wha- aim, i don't wanna shoot. (.5)
 P Leo: oh::: my god (.4)
 P Vincent: i don't gonna i'm not gonna shoot (.) are you kidding me? {switches between pointing and pointing away at Leo} (.3) do I have to? (.3) no, what if i don't. (.6) i don't wanna d(h)o this (.h) ↑ it's gonna f(h)orce me ↑
 P Leo: (h) (.h) you don't have an option. (.3)
 P Vincent: (h) i don't wanna do this (.3) I really don't wanna do this (.2)
 P Vincent [holy shit]
 P Leo: [ahh:: this is so] (.hh)
 P Vincent: {shoots} ohh:: ↓ my god. ↓ why? (..) ohh:: come o::n
 P Leo: (.hhh) (hhh)
 P Vincent: fuck this ga::me. No:: i didn't w(h)ant it to e(h)nd like this

What we see here is a familiar “mapping” of players onto their characters, despite the fact they have played a round character for the entirety of the game – putting further tension on Troisi's distinction between “objective” and “subjective” narrative positions [39]. Players refer to their characters in first person yet simultaneously reflect on their own position as players. Additionally, they use the subject pronoun “it” to refer to the agency of the game itself, clearly no longer a “game” in the sense of “free movement within a more rigid structure” as Salen and Zimmerman [40] put it, but a system of predetermined choices. We also see players rationalizing the choices made by their characters and, instead of resisting, “playing along” and roleplaying as the idiosyncratic character they are playing. One player who gets to shoot while playing the cop, Vincent, rationalizes by saying “Well, it's my job” and “there's no honor among thieves” to his co-player. At the same time, players who control the character who shoots the other discursively distance themselves from the act they engage in. One player, while having

used first and second person pronouns throughout the scene, slips into third person directly after his avatar shot the other.

P Vincent: ↑ why did he ↑ shoot im? (.) i mean i guess he-

P Leo: leo definitely was gonna kill him

P Vincent: ↑ yeah i mean ↑ he (.) had (..) all the right to

The player demonstrates what Goffman calls ‘role distancing’ [35] which pertains to the act of presenting one’s ‘self’ as being removed or at a distance from the role one is being required to play. Role distancing is one strategy which allows the individual to play the role but to resist it – for example, by keeping your eyes open when asked to pray or say grace, you communicate to the group by role distancing, that you are making no commitment to the role. As such, people get to deny “the virtual self that is implied in the role for the allocating performers” [41]. While role distancing is a well-known trait from everyday life, it has as of yet not been connected to the complex performances of streamed video game play. Let’s Players have to negotiate the space between themselves as performers vis-a-vis both the game (which fixes them into certain roles) and their potential audience (which may expect certain roles to be played). In this particular case, their linguistic behavior signals a dissonance not between their play and the narrative, but between their prior investment in the totality of their prior role and the new one they are forced to take. Taking stock of such conversations, where Glas [42] argues that “the aim with LP videos is not to create stories”, we would strongly disagree. More than “annotating their game-play in real time” [43], Let’s Players engage in all kinds of narrative life writing acts [44]. Moreover, it is the negotiation of that self that occurs in the moment that players are confronted with a forced action they do not subscribe to.

In sum, the use of role distancing in the broader narrative construction of “demonstrative play” that the Let’s Play genre consists of, shows the subtle ways in which players are claiming authorship over the uncontrollable experience that the game offers. It also demonstrates that the dissonance, here, is not primarily one between mechanics and narrative, even if we would agree upon such a binary: there is not necessarily a tension between the narrative of the betrayed partner and the mechanical consequences that flow from that. Instead, the dissonance here arises between what we could describe as the hermeneutics of the designer and that of the player.

6 Discussion

The term ‘ludonarrative’ and related vocabulary (‘video game narrative’, ‘narrative-focused game’, ‘interactive story’) has been used to describe the phenomenon of narrative in interactive media. However, clear and generally accepted definitions of what this term actually entails are elusive. Accordingly, Koenitz states that no generally accepted definition of video game narrative exists as of yet [21]. He identifies a range of positions (e.g. the experience of a video game, its content and or an analytical perspective), which clearly demonstrate the vagueness of existing terminology. In this paper we have seen how current definitions of interactive narrative are not sufficient to describe the

problems players have with the narrative of a particular game, and its narratively interpreted mechanics.

Again: the fact that we cannot conceptualize gameplay without a resort to narrative is not an attempt to police the borders of narratology and ludology. What it allows us to see is that there is a friction between the player's unfolding ideas about the characters they inhabit, and the designer's intended experience for that player. In this respect, ludonarrative dissonance describes the consequences of deeper-seated systemic tensions. In order to conceptualize this, we have proposed a different categorization by incorporating a hermeneutic approach to Koenitz' SPP model [23]. This conflict we identify lies beyond the moniker of ludonarrative dissonance: it is a disturbance in the interactive narrative experience caused by the unbridgeable distance between players' expectations (the assumptions they develop as they experience the narrative through gameplay, as shown in the *interaction and interpretation circle* in figure 1) and what is imposed by the designer, with the intention to create an engaging, immersive, and meaningful experience.

This view is supported by our analysis of Let's Play sessions as we observed players to be confused and in some cases even "rebellious" against the removal of agency, especially when the narrative that ensued subverted their conception of its integral unity. The hermeneutic strip we have introduced seemed to break, as the particular instantiation of the story turned out to differ (due to player agency) from the protostory players had imagined. This, we argue, demonstrates how players develop a sense for the level of granted agency, for instance when choosing to follow Leo's or Vincent's approach to the issues they encounter throughout the game.

As discussed above, IDN is a shared meaning-making activity between designer and audience. In the last interactive scene of *A Way Out*, shared meaning-making turns into an imposed narrative by the author. Here, the creator changes from the role of an interactive narrative designer, offering agency to the players, to the role of the traditional author who wants to convey a specific narrative meaning. Players revolt, when encountering the withdrawal of their agency as the co-creative process turns into an authored one, subverting their own co-created hermeneutical ideas about the 'protostory' (the set of all possible narratives) and the current concrete instantiation (the product created so far). This issue is related to the concept of design conventions as the conventional understanding of design methods by the audience [45]. *A Way Out*, in this context, applies the IDN design convention of "meaningful collaborative choice" throughout most of the game before finally frustrating audience's expectations when a novel, limiting interaction design is imposed.

This confusion leads to perceived dissonance, a rupture in the experience, leading to a break in the ludonarrative process. We argue that players, as a result of this interruption, instead of being invested in further narrative progression, started focusing on the change of their interaction experience – a move towards a meta-communicational level. As we know, in literature the "whole", as part of the hermeneutic movement, refers to the entirety of the story. When it comes to IDN, that whole is more than that: as the protostory it also refers to anything the player "could have done" as well. This is, in fact, what Murray [16] considers part of the transformative aspect of games: knowing

that things could have been different is constitutive for the experience, as it makes one's decisions personally meaningful.

7 Conclusion

When creating interactive digital narrative systems, designers need to find a balance between player agency and a dramatic structure to offer a consistent, meaningful narrative. The resulting set of ludonarrative experiences is an essential part of narrative-rich games that cannot be ignored. Ludonarrative dissonance can be best understood as a temporal dissonance between the player's interpretation of the ludonarrative process and the interpretation of the overarching narrative (the instantiated narrative product created so far), and a function of the player's iterative hermeneutic oscillation.

"Ludonarrative dissonance" as an umbrella term does not explain the actual dissonances that occur and which can easily be misinterpreted as creating a false dichotomy between ludic and narrative elements. Our extended SPP model is a first step towards gaining a better understanding of such dissonant experiences. The challenge that lies ahead involves identifying and explaining the actual dissonances that can occur as the construction of narrative is tightly connected to our meaning-making. To understand the complex relationship between interaction and narrative it is crucial to include a cognitive perspective in the analysis. This destabilizes the dichotomy between "ludo" and "narrative", as narrative interpretation can be situated on several dimensions of the ludonarrative experience.

Future work might focus on expanding the IDN model to include the different narrative dimensions in between which dissonance can occur. Researchers might also investigate into temporal aspects and, for instance, the dissonance between different play-through sessions (e.g. for systems that offer cross-session memory like *Save The Date*). In terms of methodology, a further exploration of Let's Play Videos as qualitative samples for the evaluation of the interactive narrative user experience is warranted – the fact that the captions of these videos can be computationally scraped is noteworthy.

A Way Out creates a collaborative experience, while highlighting the friction between collaboration and competition by which many friendships are characterized. Each player is capable of relating this "social model" to their own life. Interactive narratives, in other words, are significantly more flexible in achieving what Barthes called the writerly text, in which the reader is located as a site of the production of meaning, and for which the goal is to make the reader no longer a mere consumer, but a producer of the text [46]. Interactive Digital Narratives are not played for escapist motives alone but also for the transformative aspect: "The right stories can open our hearts and change who we are. [...] Enacted events have a transformative power that exceeds both narrated and conventionally dramatized events because we assimilate them as personal experiences." [47]

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